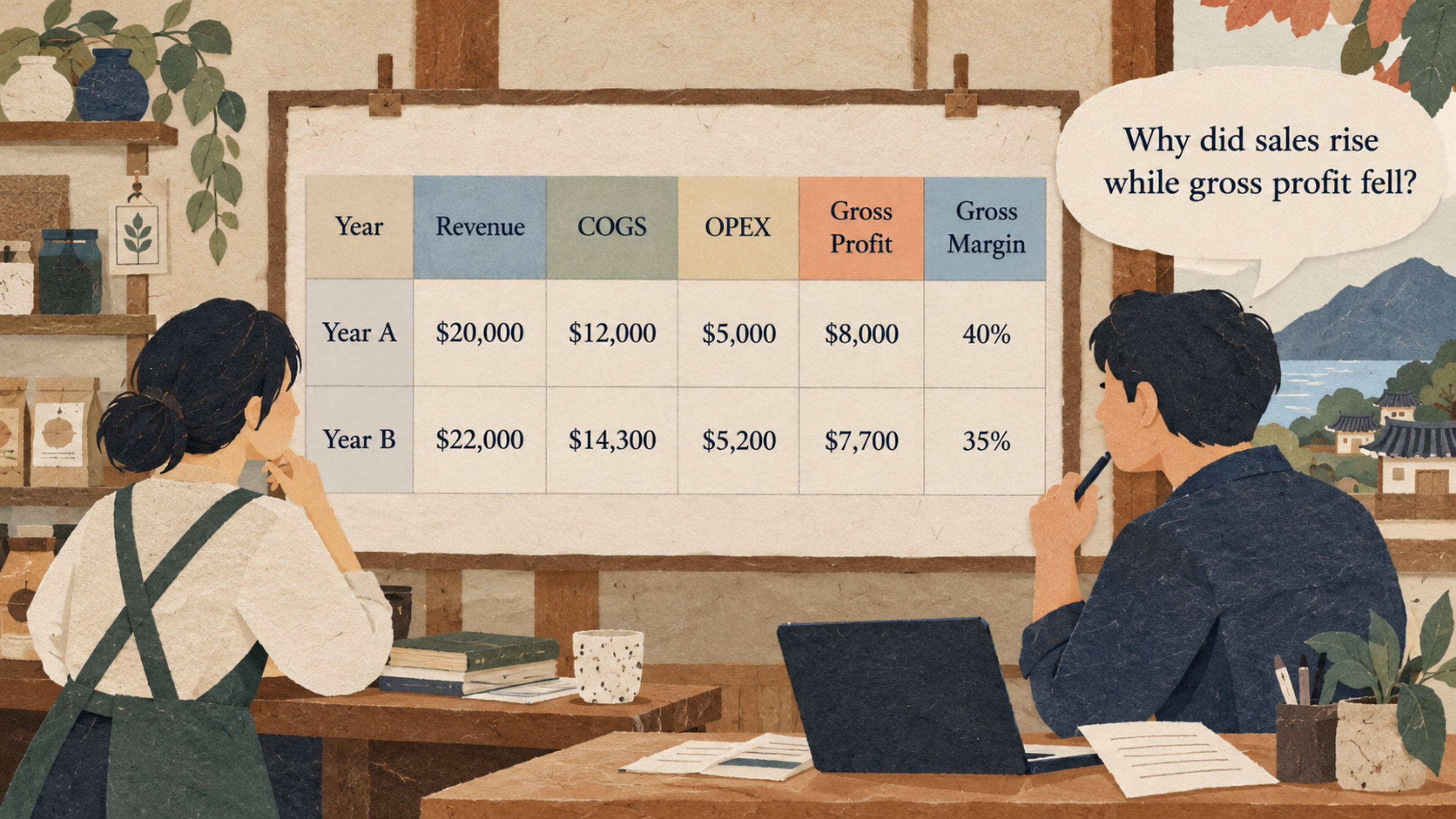




Why did sales rise while gross profit fell?

Year	Revenue	COGS	OPEX	Gross Profit	Gross Margin
Year A	\$20,000	\$12,000	\$5,000	\$8,000	40%
Year B	\$22,000	\$14,300	\$5,200	\$7,700	35%

What problem are we solving?

3 hours: 50-minute lesson, 10-minute break, 50-minute guided practice, 10-minute break, 50-minute applied practice, and 10-minute wrap-up.

SITUATION

The problem

Sales can rise while the money left after direct costs falls. A manager needs to explain the change without confusing sales, direct costs, and overhead.

TARGET

Learning goal

Calculate gross profit and gross margin, check the arithmetic, and write a short evidence-based briefing.

Practice material: Use the practice material supplied in the session Markdown.

Use these supplied facts

Do not replace missing information with guesses.

CASE MATERIAL

Practice data (fictional shop)

- Year A: revenue \$20,000; cost of goods sold (COGS) \$12,000; operating expenses (OPEX) \$5,000. Year B: revenue \$22,000; COGS \$14,300; OPEX \$5,200. COGS is the direct cost of goods sold; OPEX is the cost of running the business outside those direct costs.

BOUNDARY

Use the supplied facts

Sales can rise while the money left after direct costs falls. A manager needs to explain the change without confusing sales, direct costs, and overhead.

Concepts for this problem

Keep each concept tied to the practical decision.

UNDERSTAND

Core concepts

- Gross profit = revenue - COGS. Gross margin = gross profit / revenue x 100.
- Gross margin describes the share of sales left after direct product costs; it is not the same as final net profit.
- Separate reported figures (facts) from explanations of why they changed (hypotheses). These figures alone do not identify the cause.
- Use a calculator or spreadsheet to verify arithmetic; use AI to organize and explain checked figures.

APPLY

A simple method

- Calculate each year's gross profit and margin; compare the figures and label facts versus possible explanations; write a brief that names one question to investigate next.

What makes a prompt weak?

These examples come from the current session scenario.

AVOID

Weak example 1

Is this shop doing well? Tell me what to do.

AVOID

Weak example 2

Prove that higher sales mean the business is more profitable.

A clearer, safer prompt

Keep the source, task, constraints, and output check visible.

USE

Improved example 1

Using only the four figures for Year A and Year B below, calculate gross profit and gross margin. Show the formula and arithmetic. Do not call gross profit net profit.

USE

Improved example 2

Use these checked results: Year A gross profit \$8,000 and margin 40%; Year B gross profit \$7,700 and margin 35%. Write a 100-word manager brief: state two facts, one possible explanation labeled as unverified, and one follow-up question. Do not invent a cause.

Exercise 1: Follow the steps

Use the complete case and instructions from the session Markdown.

GUIDED PRACTICE

Exercise 1: Follow the steps

- Calculate gross profit and gross margin for both years with the formulas above. Check: Year A = \$8,000 and 40%; Year B = \$7,700 and 35%. Compare your answers with AI, then correct any mismatch yourself.

Exercise 2: Try a small variation

Use the complete case and instructions from the session Markdown.

APPLIED PRACTICE

Exercise 2: Try a small variation

- In pairs, write a short briefing for a store manager. Explain that revenue increased by \$2,000 while gross profit decreased by \$300 and margin fell by 5 percentage points. Suggest one additional piece of information to investigate; do not claim to know the cause.

How to know the work is complete

Check the output before sharing or acting on it.

SUBMIT

Deliverable

- A two-row calculation table and a 100-word manager brief with facts, an explicitly unverified explanation, and a follow-up question.

VERIFY

Success check

- Both calculations are correct, gross margin is not mislabeled as net profit, and the brief does not present a guess as a fact.